DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT

Autonomous under VTU, Belagavi

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Mini Project [BEC506]

Report On

"DESIGN AND IMPLEMENTATION OF DLDO (Digital Low Dropout Regulator)"

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Department of Electronics and Communication Engineering

CERTIFICATE

This is to certify that the mini project work entitled **"DESIGN AND IMPLEMENTATION DLDO (Digital Low Dropout Regulator)"** carried out by ADITH SORAGU (1DT23EC003), A YASWANTH (1DT23EC009), K SHOBITH KUMAR (1DT23EC040), CHALUVARAJU S N (1DT24EC401) a Bonafide student of DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT in **Bachelor of Engineering in Electronics and Communication Engineering** of the **Visvesvaraya Technological University, Belagavi** during the year 2025-2026. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Signature of the Guide

Dr. Roopa R Kulkarni

Signature of the HOD

Dr. Mallikarjun P Y

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Yours Sincerely,

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ABSTRACT

Power efficiency has been one of the most important requirements in today's electronic systems, and with the advancement of portable and battery-operated devices, LDOs need to be power efficient. Traditional analog LDOs have a lot of drawbacks, including higher quiescent current, slower transient response, and inferior efficiency for suddenly changing load conditions. In order to surmount these obstacles, this mini-project covers the design and implementation of a low-power digital LDO that can effectively increase the lifetime of batteries and reduce processor heating.

The proposed DLDO architecture integrates a high-speed comparator, digital control logic, and a PMOS power transistor array. The comparator continuously monitors the deviation between the output voltage and the reference voltage, while the digital logic dynamically adjusts the number of active PMOS switches to maintain a stable regulated output. This digitally assisted regulation significantly reduces power dissipation and improves thermal performance compared to conventional analog solutions.

Functional verification of the circuit was performed by first modeling the circuit in MATLAB/Simulink. The complete transistor-level schematic was then designed and simulated in Cadence Virtuoso. The results of transient response show stable output regulation with very minimal ripple under changing conditions of clock cycles, thereby validating robustness in this design. The overall implementation hereby confirms that the DLDO indeed performs its operation at low power and is hence suitable for modern VLSI systems, embedded processors, and portable electronics.

Results of the project is obtained the required parameters like PSRR = 20db and Quiescent Current = 2.66ma and DVFS = 2.44mw.45nm technology is used to implement the DLDO. The main agenda was to get the low quiescent current to increase the battery life and high transient response for reducing heat of the processor.

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CHAPTER I

INTRODUCTION

The increasing growth of portable, battery-driven electronics has made power management a necessary design specification. Modern processors run at a low voltage, which requires a fast, precise voltage regulation mechanism, thus making voltage regulation a critical design specification that affects battery life, with excess heat being generated when inefficiencies occur.

Digital Low Dropout Regulator (DLDO) – It is an advanced voltage regulating technique that offers a stable output voltage with less voltage drop between the input and output. Unlike the analog LDO, the DLDO uses digital control logic, comparator circuits, and digitally controlled power transistors to control the stable voltages of the LDO. This is a digital solution that improves scalability with respect to CMOS, robust voltage, temperature, and process variations.

In this project, the design and implementation of the DLDO are intended to enhance the battery life as well as prevent processor overheating. The DLDO minimizes power waste in voltage regulation by providing just enough voltage to the processor, thus making sure that there are no unnecessary power losses. The fast-transient behavior of the DLDO helps it maintain stable operation even when subjected to varying load conditions.

Accordingly, the DLDO power management solution developed has improved efficiency, increased battery life, and better thermal management, making it ideal for low-power and high-performance processor solutions.

The results demonstrate that the proposed DLDO architecture achieves stable voltage regulation, low power consumption, and effective performance across multiple voltage levels, making it suitable for low-power and DVFS-enabled applications.

Power management is a critical requirement in modern electronic systems, especially in low-power and high-performance applications such as portable devices, IoT nodes, and System-on-Chip (SoC) platforms. Among various voltage regulation techniques, **Low Dropout (LDO) regulators** are widely used due to their simplicity, low noise, and ability to operate with a small difference between input and output voltages. However, conventional **analog LDOs** face limitations in scalability, power efficiency, and adaptability when implemented in deep sub-micron technologies.

To overcome these challenges, this project focuses on the **design and analysis of a Digital Low Dropout Regulator (DLDO)**. A DLDO replaces the traditional analog control loop with digital components such as comparators, quantizers, digital controllers (PID/logic), and digitally controlled power transistors. This digital approach offers better robustness against process, voltage, and temperature (PVT) variations, improved scalability with CMOS technology, and enhanced flexibility in control.

In this project, the DLDO is modeled and verified at the **system level using MATLAB/Simulink**, where key blocks such as the error detector, quantizer, zero-order hold, and digital controller are implemented to study transient response, output regulation, and stability. Further, the design is validated at the **circuit level using Cadence Virtuoso**, ensuring realistic performance evaluation under varying load and input conditions.

The primary objectives of this project are to achieve **stable output voltage regulation**, **fast transient response**, **low output ripple**, and **high power efficiency**, making the proposed DLDO suitable for modern low-power integrated circuits.

1.1 INTRODUCTION TO DLDO

A digital LDO is a power-management circuit featuring an extremely small dropout voltage, which is the difference between the relatively higher input voltage and the output voltage for generating the stable and regulated output voltage. DLDOs are becoming a very important component in modern low- power and high-performance electronic systems that require energy efficiency, fast response time, and compact integration. Unlike conventional analog LDOs, DLDOs use comparators, digital logic, and power transistors with digitized control to regulate the output voltage.

1.2. Type of DLDO

1) **Synchronous DLDO**

Fig [1.1] shows A synchronous digital LDO uses a clock signal for the control of its digital regulation process. In this type, the output voltage is sampled and compared with a reference voltage at discrete clock intervals. Based on the comparison result, the digital controller updates the state of the power transistors only on clock edges.

Key Features:

- Requires operation of a system clock
- Predictable and stable regulation behavior
- Easier to design and verify
- Lower noise because of controlled switching instants

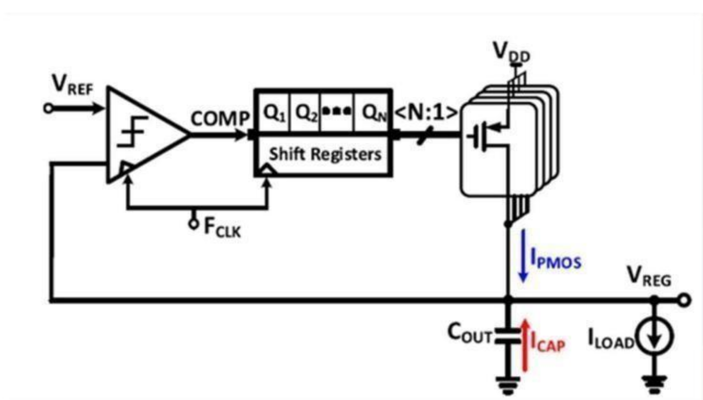

Fig 1.1: Synchronous DLDO [10]

2) **Asynchronous DLDO**

Fig [1.2] shows an Asynchronous Digital Low Dropout Regulator (DLDO), there is no need for a clock signal. The regulating process is initiated as soon as there is a change in the output voltage, with the use of comparators. The power transistors are turned ON/OFF once a voltage error is sensed.

Key Features:

- No clock required
- Extremely fast transient responses
- Reduces Latency in Voltage Correction
- Lower power consumption when accessing clocks

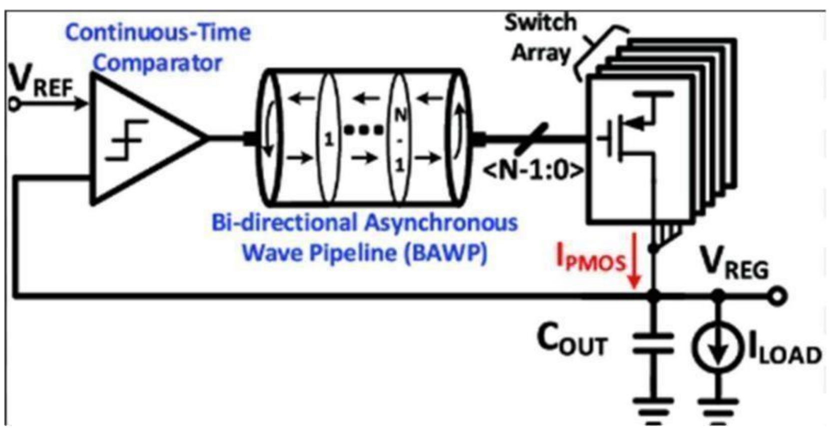

Fig 1.2: Asynchronous DLDO [5]

1.3. Use Cases Of DLDO

DLDO (Digital Low-Dropout Regulators) applications

- Numerous applications of DLDOs occur in battery-operated electronic devices like smartphones, tablets, wearables, etc.
- Use in processors and microcontrollers with integrated voltage regulation on the chip.
- Use in System-on-Chip (SoC) Designs which incorporate several independently regulated supply voltages.
- Use in Low-power, high-speed digital circuits requiring quick load transient response.
- The relevancy of the term "digital" comes from the extreme level of integration of the associated DLDOs within the scaled CMOS (Complementary Metal-Oxide Semiconductor) technology nodes.

1.4. Organization of the chapters

Chapter 1: introduction of our project, we will be using 45nm technology to implement the DLDO and we consider the parameter like the quiescent current and voltage respectively and PSRR. Introduction to DLDO, here we discuss what is DLDO and types of DLDO and there use case and also why are we using DLDO for our project.

Chapter 2: we have performed LITERATURE SURVEY and given their key takeaways through a table.

Chapter 3: It will explain the design of the system and also the overview. And the design specification and mathematical equations.

Chapter 4: It will explain the software requirement for our project. And library used in our project.

Chapter 5: In this chapter we will discuss how our project is implemented. And explains about the design of comparator and shift registers.

Chapter 6: we will discuss the results and the analysis of our project in different parameter like PSRR, DVFS, Quiescent current.

Chapter 7: we will discuss the future scope and the conclusion of our project.

1.5. Summary

A Digital Low Dropout Regulator (DLDO) is a power management IC that is capable of delivering a stable output voltage with extremely low dropout voltage, making it ideal for use in modern power- sensitive and high-performance systems. Unlike the analog LDO, the DLDO employs digital components such as comparators, logic, and digital power transistors, which make it faster with improved power efficiency.

The types of DLDOs are generally categorized into two: synchronous DLDOs and asynchronous DLDOs. The synchronous DLDOs generally work with a clock signal, providing stable voltage with less noise, whereas asynchronous DLDOs work without a clock signal, which provides a faster transient when compared with a synchronous DLDO, consuming less power because of which DLDOs are commonly used in battery-driven devices, processors, microcontrollers, SoCs, and high-speed low- power digital circuits.

CHAPTER 2

LITERATURE SURVEY

In [1] This is proposed architectural advancements for digital low-dropout (DLDO) regulators. In the paper, new design methodologies are presented to enhance transient response and power efficiency of DLDO. It details enhancements in modern circuit architectures that can boost regulators' performance several folds for low-voltage digital systems. This study speaks about improvement in the control mechanisms and improvement of loop stability, thereby making the DLDO more suitable for next-generation power management systems.

In [2], Xue et al. present a 0.5-V input digital LDO implemented in 65 nm CMOS technology with high efficiency of 98.7% and very low quiescent current of 2.7 µA. This work presented here has demonstrated that even under ultra-low-power supply conditions, the proposed DLDO achieves high efficiency suitable for battery-powered devices. The experimental results depicted in the work have also demonstrated significant improvement in load regulation and response time, thereby making the design suitable for portable electronic circuits.

In [3], the authors had proposed a low-quiescent-current asynchronous digital LDO with PLL-modulated Fast-DVS power management that was designed for 40 nm SoC applications. Their focus is to enhance dynamic performance for high-speed processors but at minimum static power consumption. The results show remarkable enhancements in response speed, with reduced voltage droop, hence giving better performance to the MIPS-based systems.

In [4], the authors are presenting various low-power design techniques and their implementation strategies in VLSI circuits. These methods include supply voltage scaling, multi-threshold CMOS, optimization of transistor sizing, and leakage reduction. The research by the authors shows how all these methods are useful in realizing digital.

In [5] introduce some methods of LCO reduction in DLDO regulators. This is considered a major factor in voltage stability and output ripple. In this paper, an improved control strategy has been proposed to enhance transient response and reduce oscillations due to varying load conditions. The simulation results showed a significant reduction in ripple amplitude, thereby allowing DLDOs to become more efficient for high-performance low-voltage systems such as SoCs and mobile processors.

In [6], the authors review trends and development perspectives of DLDO voltage regulators for low- voltage mobile applications. This review covers challenges in efficiency, quiescent current reduction, transient response, and output ripple management. The paper stresses recent advances in control architectures, in power- management integration, and the need to optimize DLDOs for fast dynamic power scaling. It also summarizes future research opportunities in energy-efficient mobile and wearable electronics.

In [7], an asynchronous Digital Low-Dropout (DLDO) regulator with a dual adjustment mode is proposed, suitable for ultra-low voltage input applications. The document describes the architecture of how transient response and power efficiency are improved by the dual-mode control operation.

In [8], a Digital Low-Dropout regulator is proposed, whose control mechanism is based on a Voltage- Controlled Oscillator. The presented approach, if compared with classical approaches, improves fast transient response and stability for changing load conditions. The proposed VCO-based DLDO has shown substantial improvement in output ripple and settling time, thus proving to be suitable for high- speed applications like advanced SoC power delivery systems.

In [9] In this context, a high-performance DLDO with dithering control and dynamic frequency scaling was proposed exclusively for IoT applications in. Advantages regarding efficiency and output ripple reduction are described in the case of large changes.

In [10] This is an open-source project published on GitHub that deals with the design of a digital LDO (DLDO) for power-efficient low-voltage SoC voltage management. In this paper, the work is oriented to solve the output voltage ripple due to LCO and perform DVS using a DLDO architecture.

2.1 LITERATURE SURVEY TABLE

Reference number	Year	Methodology	Performance Parameter	Result Achieved
[1]	2010	Proposed improved DLDO architectures featuring enhanced digital control and loop stability	Output Voltage Ripple, Load Regulation, Area Overhead	Transient response improved by **>30%**, output voltage ripple reduced by **~25%**, efficiency improved by **~10%**
[2]	2015	Implemented ultra-low-voltage DLDO using digital control logic in 65-nm CMOS	Line Regulation, Static Power Consumption, Dynamic Power Consumption, Leakage Power	Input voltage: **0.5 V**, current efficiency: **98.7%**, quiescent current: **2.7 µA**
[3]	2016	Designed asynchronous DLDO with PLL-based fast dynamic voltage scaling	Dynamic Power Reduction, Leakage Power Reduction, Thermal Power Dissipation	Voltage droop reduced by **~40 mV**, settling time < **100 ns**, quiescent current **< 5 µA**
[4]	2016	Applied voltage scaling, MTCMOS, transistor sizing, and leakage control	Current Efficiency (%),Quiescent Current (Iq), Transient Response Time, Power Dissipation	Power consumption reduced by **30–50%**, leakage power reduced by **~40%**
[5]	2017	Enhanced digital control to suppress limit cycle oscillations (LCO)	Quiescent Current (Iq), Power Dissipation	Output ripple reduced by **>50%**, transient recovery time improved by **~35%**
[6]	2017	Comprehensive review of state-of-the-art DLDO architectures and challenges	Current Efficiency (%),Dropout Voltage, Dropout Voltage, Load Regulation, Line Regulation, Area Overhead	DLDO efficiency up to **99%**, quiescent current as low as **1–5 µA**
[7]	2020	Dual-mode asynchronous control for ultra-low-voltage DLDO	Output Voltage Ripple, Limit Cycle Oscillation (LCO)	Load regulation improved by **~20%**, transient response **< 80 ns**, no increase in quiescent current
[8]	2021	Used voltage-controlled oscillator (VCO) for digital feedback control	Transient Response Time, Power Dissipation	Settling time reduced by **~45%**, output ripple **< 10 mV**
[9]	2023	Adaptive DLDO using dithering control and dynamic frequency scaling	Current Efficiency (%),Quiescent Current (Iq), Dropout Voltage	Efficiency **>95%**, ripple reduced by **~30%**, quiescent current **< 3 µA**
[10]	2023	DLDO implemented with DVS and LCO mitigation using Verilog-A	DVFS, PSRR	Output ripple reduced by **~20– 30%**, stable operation under dynamic loads

2.2 Problem Definition

Modern battery-driven electronic devices, such as mobile processors, demand efficient voltage regulation. Low Dropout Regulators have efficiency limitations at low currents, a slow response, a higher power loss, reduced battery life, and result in excessive heating of the processor. Processors have a dynamic workload, changing instantaneously, requiring efficient voltage. When inefficient, the voltage causes an energy loss in the form of heat. This affects the reliability of the device, creating a poor user experience.

The task is to design and realize a Digital Low Dropout Regulator (DLDO) for adaptive regulation of the supply voltage with high efficiency and fast transient time and to minimize the power loss. The DLDO needs to be dynamic to the changes in the load conditions of the processor and should minimize the voltage variations to increase the battery life and reduce the thermal stress on the processor. All the above needs to be achieved with stability and low overhead cost and compatibility with modern CMOS technology.

2.3 Objectives

The purpose of the proposed Digital Low Dropout Regulator (DLDO) is to perform efficient and dynamic voltage scaling in processor core voltage regulation for battery-driven systems in an attempt to improve battery life and suppress the heating of processors. The proposed DLDO seeks to achieve minimal power losses in the voltage scaling process by promoting high current efficiency and low dropout voltage to fully utilize the battery's energy for an extended period of time. The proposed design minimizes power losses and reduces the quiescent current as it operates in the idle mode by reducing the power losses. The aspect of dynamic voltage scaling in the design enables the regulator to dynamically scale the supply voltage based on the processor's workload to suppress the unnecessary dynamic power losses associated with the system's processor during voltage scaling to improve energy efficiency.

2.4 Summary

In this chapter, a systematic literature survey on Digital Low Dropout Regulators (DLDOs) has been discussed, covering its application in increasing power efficiency, increasing battery life, and preventing heating of processors in state-of-the-art battery- driven systems. The surveyed literature reveals that there are immense improvements in DLDO design methods, encompassing improved digital methods, asynchronous design, dual-mode, PLL & VCO-based control, and limit cycle oscillations (LCO). These methods altogether aim to provide high current efficiency, low quiescent current, rapid transient recovery, minimized output voltage ripple, and improved load & line regulations. Moreover, there has been a growing emphasis on applying DVS & low- power VLSI design methodologies to design power-efficient & thermally stable processors. Finally, to meet identified shortcomings in existing DLDO design, a set of precise goals have been defined to provide adaptive, stable, and low-loss voltage scaling, that are compatible with existing CMOS processes & aim to optimize battery life, avoiding heating effects in processors.

CHAPTER 3

DESIGN OF THE SYSTEM

3.1. Overview of the System

The proposed system is centered on designing and implementing a Digital Low Dropout Regulator (DLDO) that is intended to increase power efficiency, prolong the life of the battery, and prevent the processor from overheating in a low voltage digital system. The DLDO is supposed to produce a stable supply voltage with a fast transient response under dynamic loading.

In this particular project, the design and implementation of the DLDO are carried out with the help of Cadence Virtuoso on a circuit level. The fundamental components, including the comparators, logic, power transistors, as well as feedback loops, are implemented on a transistor level with the use of CMOS. Cadence Virtuoso helps facilitate the analysis of interactions between the analog and digital systems, thereby making it possible to assess the most essential properties such as voltage stability, transient, current, and power.

To effectively support the design approach at the circuit level, the design tool of choice for modeling at the system level is MATLAB Simulink, which simulates the behavior of the DLDO. It is a high-level design environment that facilitates the modeling of the control system architecture, system-level analysis, and verification of the control design at the system level. The design tool therefore supports design verification at the system level.

The proposed solution, which integrates system-level simulation within the environment of MATLAB Simulink with circuit design with Cadence Virtuoso, ensures functional correctness as well as feasibility with respect to the DLDO design. This two-step solution provides a highly efficient, reliable, and scalable solution for power management, particularly for low-power systems as well as high- performance systems.

The DLDO minimizes power dissipation by operating with a **very small dropout voltage** and eliminating continuous analog bias currents. Power loss is primarily determined by the product of dropout voltage and load current, making the DLDO particularly efficient for low-voltage.

3.2. Design of System level model using MATLAB SIMULINK

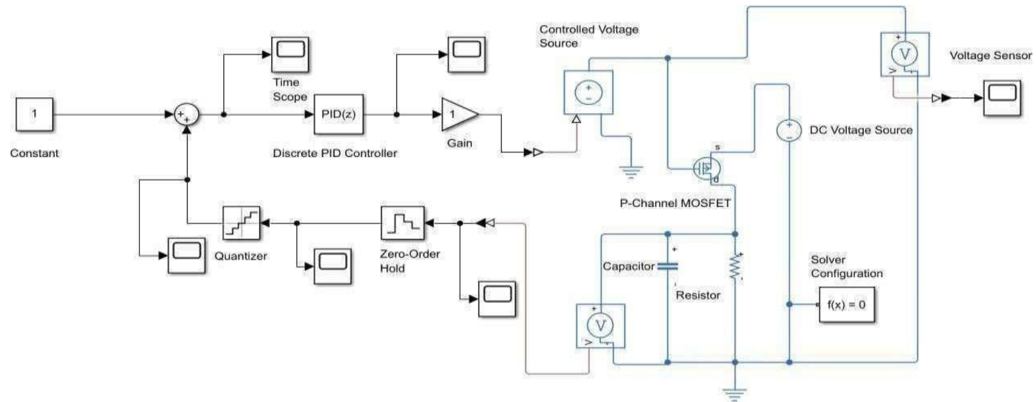

Figure 3.1 MATLAB Schematic of DLDO

Explanation of the Component in DLDO System in fig [3.1].

1. Reference / Constant Block

- **Function:** Provides the desired reference voltage (Vref) for regulation.
- **Specification:**
  - Reference Voltage (Vref): **0.8 V – 1.2 V** (processor core voltage range)
  - Type: Constant DC value
- **Design Objective:** Acts as the target voltage that the DLDO must maintain at the output.

2. Summation (Error Generator)

- **Function:** Computes the error between reference voltage and sensed output voltage.
- **Specification:**
  - Error = Vref − Vout
  - Continuous comparison at each sampling instant
- **Design Objective:** Enables feedback control for voltage regulation.

3. Discrete PID Controller

Expression of PID:

$$P + I \cdot T_s \frac{1}{z-1} + D \frac{N}{1 + N \cdot T_s \frac{1}{z-1}}$$

Where:

Proportional (P): 1

Integral (I): 1

Derivative (D): 0

Filter coefficient (N): 100

- **Function:** Generates digital control signal to regulate MOSFET gate voltage.
- **Specification:**
    - Controller Type: **Discrete PID**
    - Sampling Time (Ts): **10–100 ns**
    - Proportional Gain (Kp): Selected for fast response
    - Integral Gain (Ki): Minimizes steady-state error
    - Derivative Gain (Kd): Improves stability and transient response
- **Design Objective:** Ensures fast transient response with minimal overshoot and stable operation.

4. Quantizer

- **Function:** Converts the continuous PID output into discrete digital levels.
- **Specification:**
    - Quantization Step Size: Based on allowable voltage ripple (≈ 1–5 mV)
- **Design Objective:** Mimics digital control behavior of DLDO and limits resolution-induced noise.

5. Zero-Order Hold (ZOH)

- **Function:** Holds the quantized control signal constant during each sampling interval.
- **Specification:**
    - Sampling Time: Same as PID controller
- **Design Objective:** Represents digital-to-analog behavior in discrete-time DLDO control.

6. Gain Block

- **Function:** Scales the digital control signal to an appropriate gate-drive level.
- **Specification:**
  - Gain Value: Tuned to match MOSFET gate threshold (typically 1–5)
- **Design Objective:** Ensures correct gate voltage swing for MOSFET operation.

7. Controlled Voltage Source

- **Function:** Converts control signal into analog voltage applied to MOSFET gate.
- **Specification:**
  - Output Range: **0–Vin**
  - Response Time: Fast (negligible delay)
- **Design Objective:** Acts as a digital-to-analog interface in the DLDO.

8. P-Channel MOSFET (Pass Transistor)

- **Function:** Regulates output voltage by controlling current from input supply.
- **Specification:**
  - Type: P-Channel MOSFET
  - On-Resistance (Rds(on)): Low (mΩ range)
- **Design Objective:** Acts as the pass element with low dropout voltage and high efficiency.

9. Input DC Voltage Source

- **Function:** Represents the battery or supply voltage.
- **Specification:**
  - Input Voltage (Vin): **0.8- 0.85**
- **Design Objective:** Supplies power to the DLDO for regulation.

10. Output Capacitor

- **Function:** Filters output voltage ripple and improves transient response.
- **Specification:**
  - Capacitance: **10 µF – 50 µF**
- **Design Objective:** Reduces output ripple and enhances stability.

11. Load Resistor (Processor Load Model)

- **Function:** Represents processor load current.
- **Specification:**
    - Resistance Range: **10 Ω – 50 Ω**
    - Load Current: Variable (light to heavy load)
- **Design Objective:** Models dynamic processor current demand.

12. Voltage Sensor

- **Function:** Measures output voltage for feedback.
- **Specification:**
    - Accuracy: High
- **Design Objective:** Provides accurate feedback for closed-loop control.

13. Time Scope / Display Blocks

- **Function:** Visualize signals such as output voltage and control signals.
- **Specification:**
    - Used for analysis only
- **Design Objective:** Observe transient response, ripple, and stability.

14. Solver Configuration

- **Function:** Defines simulation parameters.
- **Specification:**
    - Solver Type: Discrete / Fixed-step
    - Step Size: Matches sampling time
- **Design Objective:** Ensures accurate simulation of digital control behavior.

3.3 Design of DLDO using Cadence Virtuoso

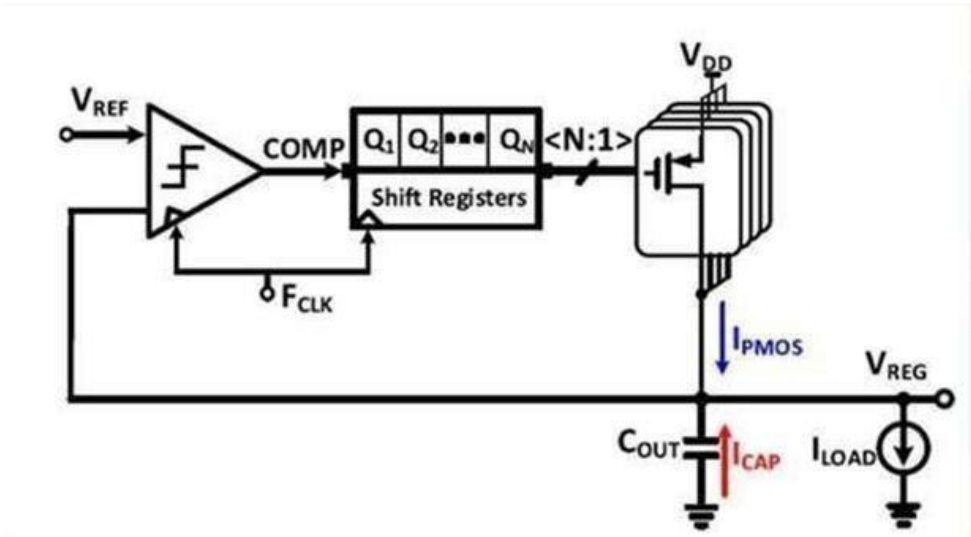

Fig 3.2 DLDO Design used in Cadence [10]

Fig [3.2] shows the proposed DLDO architecture consists of a voltage comparator, digital control logic (shift registers), an array of PMOS pass transistors, and an output filtering stage. Each block is designed to ensure efficient voltage regulation, fast transient response, and low power dissipation for battery-operated processors.

1. Reference Voltage (VREF)

- **Function:** Defines the desired regulated output voltage.
- **Specification:**
  - Voltage Range: **0.6 V – 1.2 V**
  - Accuracy: ±1%
- **Design Intent:** Acts as the target voltage against which the output voltage is continuously compared.

2. Voltage Comparator (COMP)

- **Function:** Compares the sensed output voltage (VREG) with VREF.
- **Specification:**
    - Type: Dynamic / clocked comparator
    - Power Consumption: Ultra-low
- **Design Intent:** Generates a digital error signal indicating whether VREG is above or below VREF.

3. Clock Signal (fCLK)

- **Function:** Synchronizes digital control operations.
- **Specification:**
    - Frequency: **25MHz**
    - Duty Cycle: 50%
- **Design Intent:** Controls the speed of voltage regulation and transient response.

4. Digital Control Logic – Shift Registers

- **Function:** Converts comparator output into a thermometer-coded digital control word.
- **Specification:**
    - Control Type: Incremental / Decremented
- **Design Intent:** Enables fine-grained adjustment of the number of active PMOS devices, reducing
- limit cycle oscillations.

5. PMOS Pass Transistor Array (Q_1–Q_n)

- **Function:** Supplies regulated current from VDD to the load.
- **Specification:**
    - Transistor Type: PMOS
    - Unit Transistor Width: Optimized for current step size
    - Total Drive Current: Matches peak load demand

- **Design Intent:** Provides digitally controlled current scaling while maintaining low dropout voltage.

6. Input Supply Voltage (VDD)

- **Function:** Main power source (battery).
- **Specification:**
  - Voltage Range: **1.0 V – 1.8 V**
- **Design Intent:** Supplies power to the DLDO and load.

7. Output Capacitor (COUT)

- **Function:** Filters voltage ripple and stabilizes output.
- **Specification:**
  - Capacitance: **0.5 µF – 10 µF**
  - ESR: Low
- **Design Intent:** Reduces output voltage ripple and improves transient response.

8. Load Current (ILOAD)

- **Function:** Represents processor core current demand.
- **Specification:**
  - Current Range: **10uf-50uf**
  - Dynamic Behavior: Rapid load changes
- **Design Intent:** Emulates real processor operating conditions.

9. Regulated Output Voltage (VREG)

- **Function:** Supplies stable voltage to the processor.
- **Design Intent:** Ensures reliable processor operation with minimal heat generation.

3.4. Summary

For system-level modeling, design verification, and validation of the DLDO, MATLAB Simulink has been used. The tool played a significant part in interpreting the dynamic performances of the regulatory element by assessing transitory responses, stability, and control capabilities against varying loads. The logic for control, feedback, and the power part of the design have been modeled at a high level.

The circuit-level design and implementation of the DLDO were performed with the use of the Cadence Virtuoso tool. The design of critical blocks such as the comparator, logic, power MOS transistors, and the output capacitor was carried out at the transistor level. The simulations offered by the Cadence tool delivered accurate results concerning performance metrics such as voltage regulation, transient, and power efficiency.

CHAPTER 4

SOFTWARE REQUIREMENT

4.1. MATLAB / Simulink

MATLAB is a high-level computational software used for system modeling, simulation, and analysis. In this project, MATLAB/Simulink is used to develop the functional-level model of the Digital Low Dropout Regulator (DLDO). The comparator, digital control logic, and PMOS switching network are represented using Simulink blocks to verify the behavior of the regulator before hardware-level implementation.

Using MATLAB helps ensure that the DLDO responds correctly to variations in reference voltage, load conditions, and control signals. This early validation reduces design errors and confirms the stability and regulation capability of the system.

4.2. Cadence Virtuoso

Cadence Virtuoso is an industry-standard electronic design automation (EDA) tool used for transistor-level design, simulation, and verification. In this project, Cadence is utilized to implement the complete DLDO circuit using PMOS pass transistors, comparator symbols, and digital control blocks.

The Spectre simulator in Cadence performs accurate transient and electrical analysis, allowing measurement of output voltage regulation, ripple, switching behavior, and overall power efficiency. This tool ensures that the DLDO meets practical low-power design requirements and behaves correctly under real circuit conditions.

4.3 Key Features of Cadence Virtuoso

1. Virtuoso Schematic Editor Used to draw the transistor-level schematic of:
   - Pass element of P-Mos’s array
   - NOT GATE
   - Resistance and capacitance(load)

2. Spectre Analog Simulator

Spectre was used for all circuit simulations:

- DC Analysis: For assessing regulation and dropout behavior.
- Transient Analysis: Verification of settling time, overshoot/undershoot, and dynamic behavior.

4.4 Library Used in this project

1. Technology (PDK) Library

- **Library:** gpdk45
- **Purpose:**
  - Provides MOSFET models (NMOS, PMOS)
  - Provides analog circuits
- **Usage in DLDO:**
  - PMOS pass transistor
  - Digital logic gates

2. AnalogLib

- **Provided by:** Cadence (default)
- **Purpose:**
  - Ideal analog components
- **Common Cells Used:**
  - vdc – DC voltage source
  - vpulse – clock signal
  - res – resistor
  - cap – capacitor
  - gnd – ground

4. DigitalLib / logic library (if used)

- **Purpose:**
  - Digital control blocks (optional)
- **Usage in DLDO:**
  - Counters
  - Flip-flops
  - Comparators

- Control logic for PMOS array

5. HdlLib (Optional)

- **Purpose:**
  - Behavioral modeling using **Verilog-A**
- **Usage in DLDO:**
  - Digital controller
  - Comparator modelling

4.4 Summary

The DLDO regulator was designed and simulated using **Cadence Virtuoso** with the **GPDK 45 nm technology library (gpdk45)** for implementing MOSFET devices and following technology-specific design rules. The **analogLib** and **basic** libraries were used to include voltage sources, passive components, ground, and schematic pins required for circuit-level simulations.

In addition, **MATLAB/Simulink** was used for **system-level modeling and verification** of the DLDO, enabling analysis of output voltage regulation, transient response, and control behavior before circuit implementation. The combined use of MATLAB/Simulink and Cadence Virtuoso ensured accurate design validation from system level to transistor level.

CHAPTER 5

IMPLEMENTATION

5.1 Flowchart of DLDO

Given flow chart fig [5.1] represents a complete low-power digital circuit design flow, starting from system specification and ending with circuit implementation and Evaluating Analysis. It illustrates how a design progresses through architectural planning, system-level modeling, RTL and circuit design, power optimization, and verification using industry-standard tools.

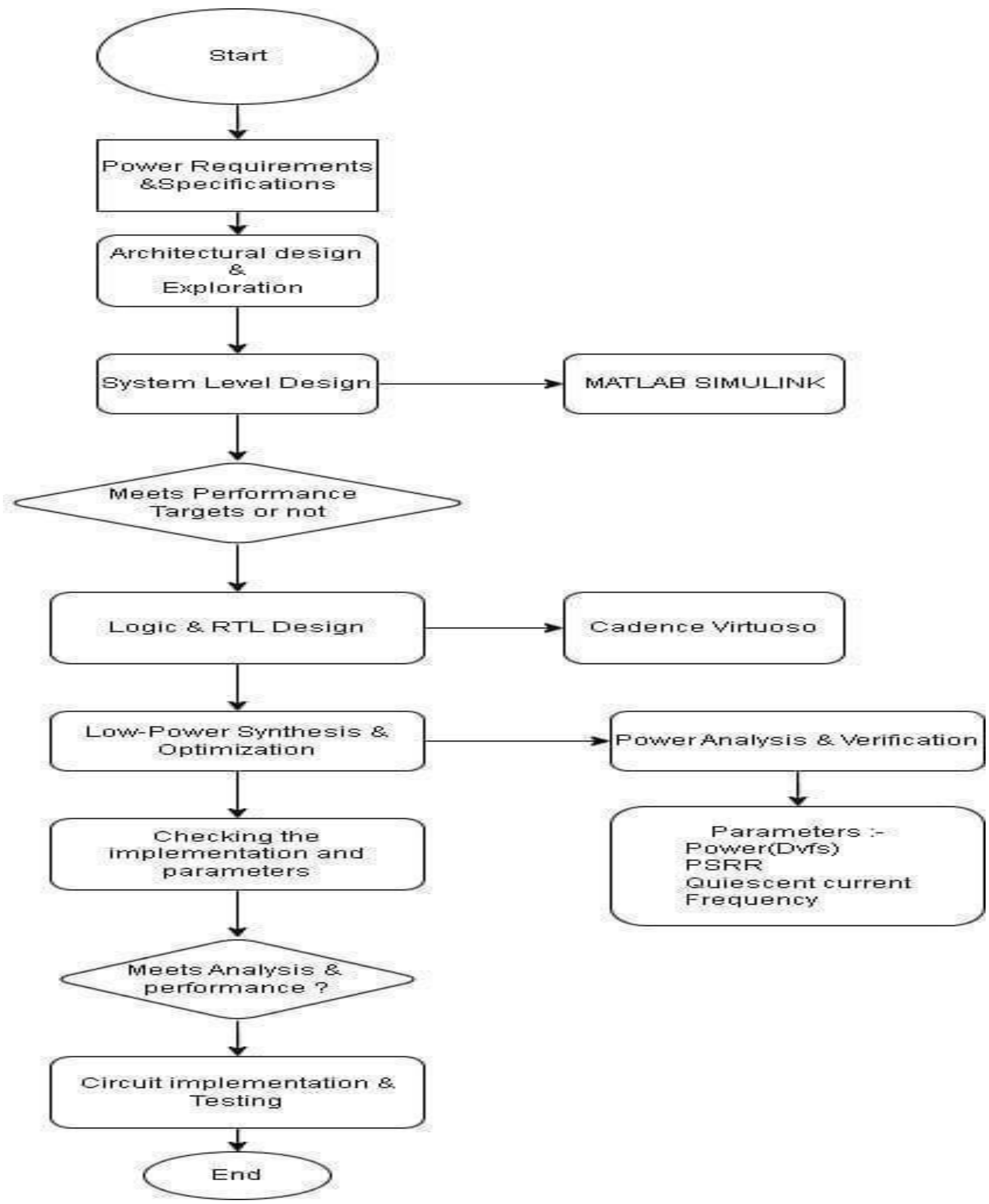

Fig 5.1: Flowchart Of DLDO

The flowchart shows fig [5.1] the design and verification process of a low-power digital system (for example, a DLDO) from the concept stage to the actual hardware stage.

The very first step to do is defining a device by determining its power requirements and writing specifications such as voltage levels, load, and performance constraints. Voltage reference(vref) =1V, 0.8V, 0.6Va, 0.4V ,0.2V. Voltage Input (vin) =1V, 0.8V, 0.6V, 0.4V, 0.2V. Frequency(f)=10Mhz.and Parameters like Quiescent Current, PSRR, Settling time, Overshoot time.

The immediate next steps after a set of requirements are architectural design and exploration. The goal of this process will be to produce an architectural design that would have an optimum balance of power, performance, and area. Architectural Design consists of Three main blocks Comparator, Shift Register, P-MOS array.

At the system-level design stage, behavior modeling, and simulation of the system with MATLAB Simulink are performed to verify the dynamic performance. The design performance will be sub- targets; hence, the architecture and parameters will be changed accordingly. System Level Modelling consists of Quantizer, Controlled Voltage Source and PID Controller. Voltage reference(vref) = 1,0.8,0.6,0.4,0.2, Resistance (Kohms) = 10,20,30,40,50 and Capacitance(uf) = 10,20,30,40,50.

The next step after obtaining system-level targets is logic and RTL design, local and by circuit-level confirmation with Cadence Virtuoso. Here RTL code id used to create the blocks which were used in architectural design.

Low-power synthesis and optimization techniques are still being employed to keep the design energy-efficient. Techniques were used DVFS (Dynamic Voltage Frequency Scaling) for the Power calculated in circuit design.

Presently, power analysis, and verification are being done alongside the main parameters recording such as power dissipation, PSRR, quiescent current, and operating frequency.

Besides, the implementation and the measured parameters are being very thoroughly checked for their correctness.

In case the design operates as per the analysis and performance requirements, it may be allowed to go to the stage of circuit implementation and testing, thus confirming operation in reality. workflow is complete with the design is successful.

5.2 Comparator Design

We have designed Comparator using the RTL code. We have used Verilog A for the coding part. We have used Verilog A because it is an analog hardware description language (HDL) used to model the high-level behavior of analog circuits, components, and networks. From the code we have created the symbol that can be used in Cadence Virtuoso.

Comparator block:

The figure shows the clocked comparator (clkcomp) block to be incorporated in the proposed Digital LDO / control system as seen in the Cadence tracer by SimVision.

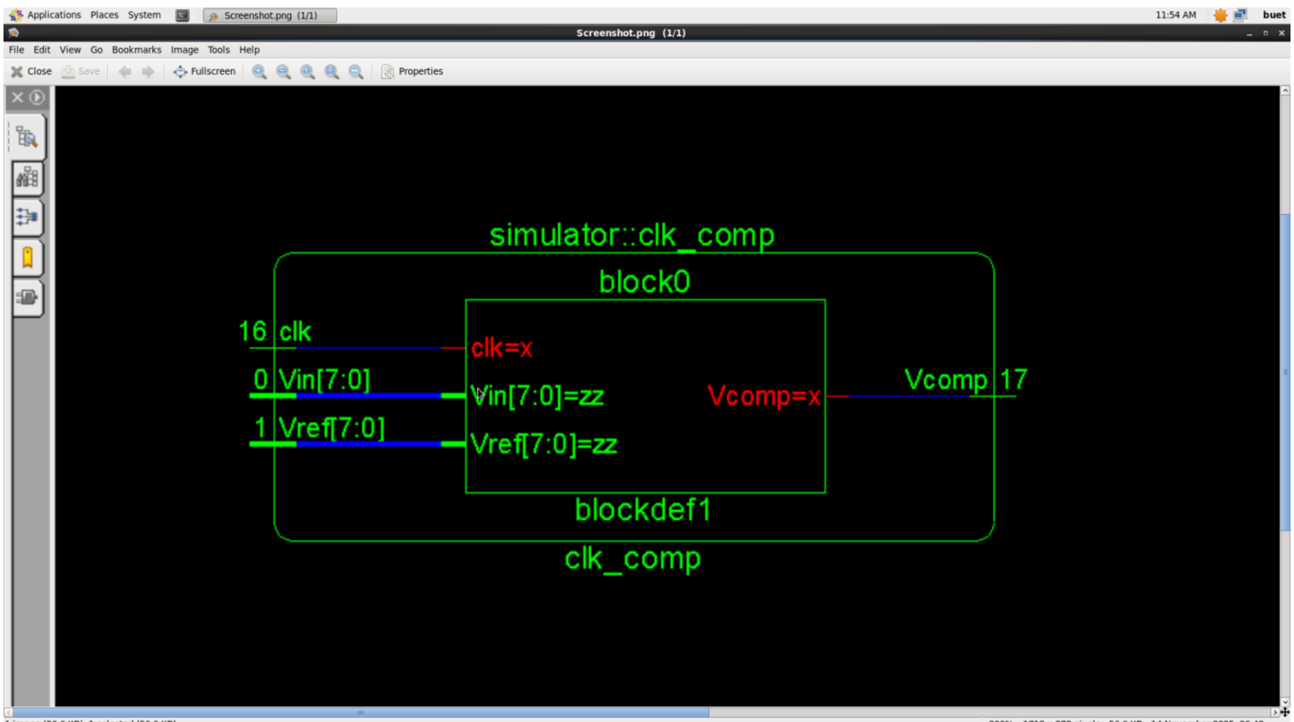

Fig 5.2 Block Diagram of Comparator

In fig [5.2] shows a 16 clock signal (clk) is applied to the block and causes the comparison operation to be performed between input voltage signal Vin [7:0] and the reference voltage signal Vref[7:0]. Both Vin and Vref have been modeled as 8-bit digital signals, which implies that the design uses more of a digital comparison mode than an analog comparator. The comparison within clkcomp module is only done on active clock edges hence saves unnecessary switching activity preventing power waste.

The comparator produces an output signal Vcomp, which is indicated with a figure of 17 effective meaning the digital comparison result between the difference between Vin and Vref. The digital control logic also uses this output to vary the proportion of pass elements/switching array of the regulator. In general the schematic brings to fore a clock-driven comparison scheme, synchronously, so that the operation can be stable, less sensitive to noise and provide stable voltage regulation over varying operating points, appropriate to DVFS-enabled low-power VLSI systems.

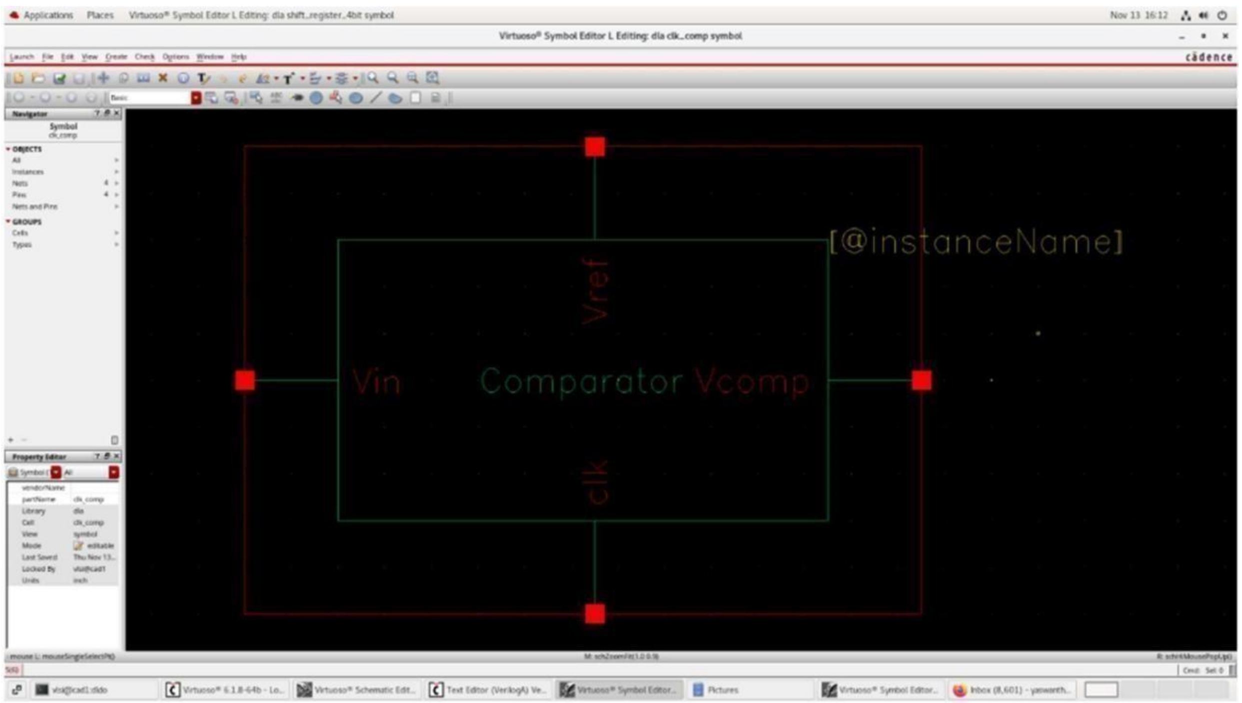

Fig 5.3: Comparator Symbol using RTL Code

This figure [5.3] shows a clocked comparator (clk comp) designed in Cadence Virtuoso Symbol Editor that will be used in the hierarchical integration in the complete Digital LDO system. This symbol gives a clean and abstract view of the comparator block and this enables it to be easily instantiated at higher design levels without revealing the internal circuitry.

The symbol has four primary outlets: Vin, Vref, clk and Vcomp. Vin terminal is the input voltage (or a digital approximation) sampled at the regulator output and Vref is the reference voltage, which specifies the desired regulated level. clk input is a system clock, which provides synchronization to the comparison operation whereby the comparator will only evaluate Vin and Vref with specific clock edges which can

Be used to reduce the sensitivity of noise interference and avoid unwanted switching activity. The output of the Vcomp is the comparison CTL which is used by the digital control logic to push the pass transistor array or switching network of the LDO.

A very clear functional block is established by the rectangular shape of the symbol, which facilitates modular and re-usable design approach. The existence of instance name placeholder allows instantiating the comparator multiple times in bigger systems when it is necessary.

5.3 Shift Register Design

We have designed Shift Register using the RTL code. We have used Verilog A for the coding part.We have used Verilog A because it is an analog hardware description language (HDL) used to model the high-level behavior of analog circuits, components, and networks. From the the code we have created the symbol that can be used in Cadence Virtuoso. In this project, a 4-bit Serial-In Serial-Out (SISO) shift register is designed using Verilog-A. The purpose of using Verilog-A is to represent the digital shifting behavior in an analog simulation environment, allowing seamless integration with analog front-end circuits, ADCs, DACs, clock generators, or custom analog logic interfaces.

Shift Register:

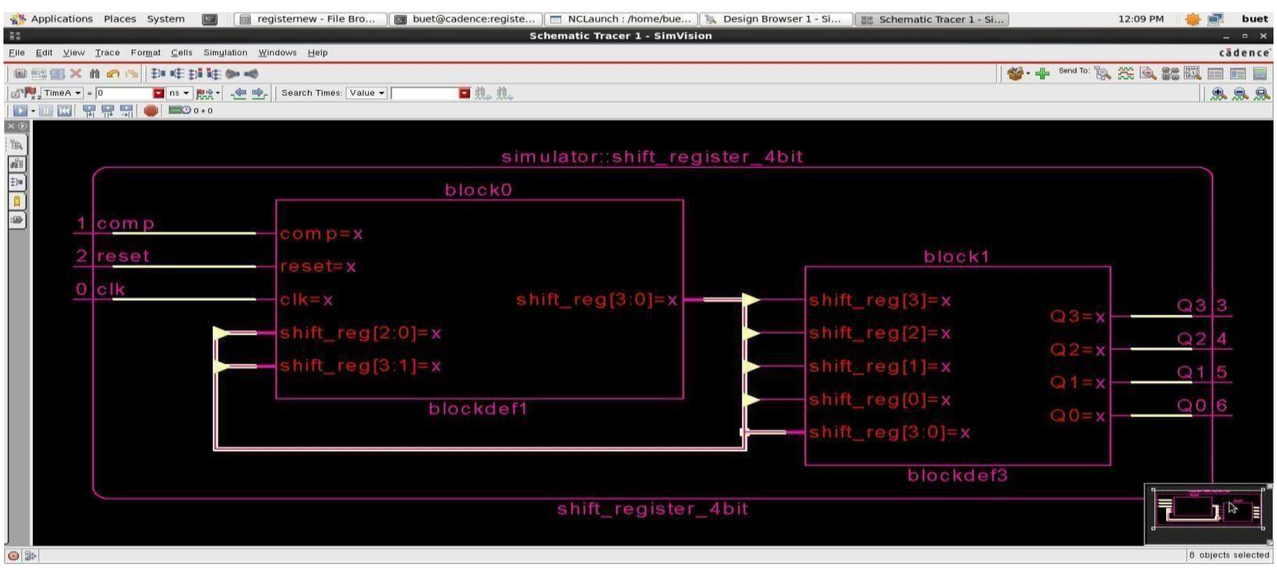

Fig 5.4: Block Diagram of Shift register (SISO)

In fig [5.4] the simulation level schematic representation of a 4-bit shift register (shiftregister4bit) as seen in Cadence SimVision Schematic Tracer is shown in the figure. This block constitutes a critical component of a digital control logic that will be employed in the presented regulator system, and sequential data shifting will be necessary in this regard to propagate the decisions and generate control signals. The shift register uses three main inputs, which are comp, reset and clk. The comp signal is the serial input, which is usually the output of the comparator, and the decision data that must be stored and shifted with clock cycles. The clk line is used to synchronize the shifting operation which only allows data movement when the clock edge is in the active state. The reset line clears the shift register, setting all the bits stored there to some known state, and ensuring that the startup can be determined. The design itself is internalized into two logic blocks. Block0 does the core shifting operation in which the incoming comp value is loaded into this least significant bit and the current contents of the register are shifted to higher bit positions (shiftreg[3:0]). Block1 is the output stage, which transforms the contents of the internal shift register to the external outputs Q0-Q3. These productions reflect the history in comparator decision making between multiple clock cycles. The schematic display reveals the intermediary signals including, but not limited to, shiftreg[3:0] and individual bit lines, which illustrates the data flow sequence in the register. In general, this 4-bit shift register can be used to filter and stabilize the comparator output over time reducing variations due to noise and enhancing the stability of user-controllable decisions.

Shift Register Symbol:

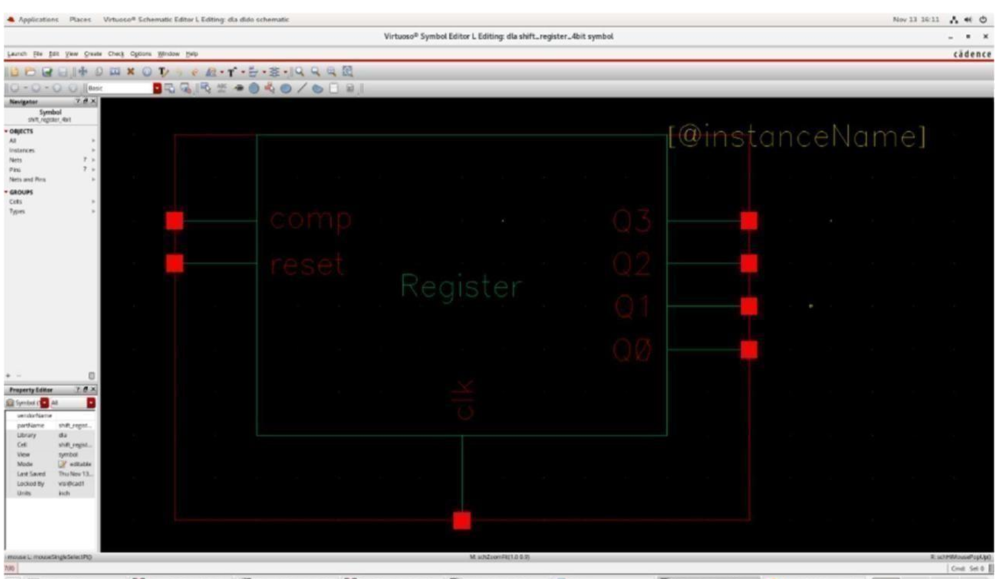

Fig 5.5: Shift Register Symbol using RTL code

5.4 Cadence Virtuoso Transistor-Level Design

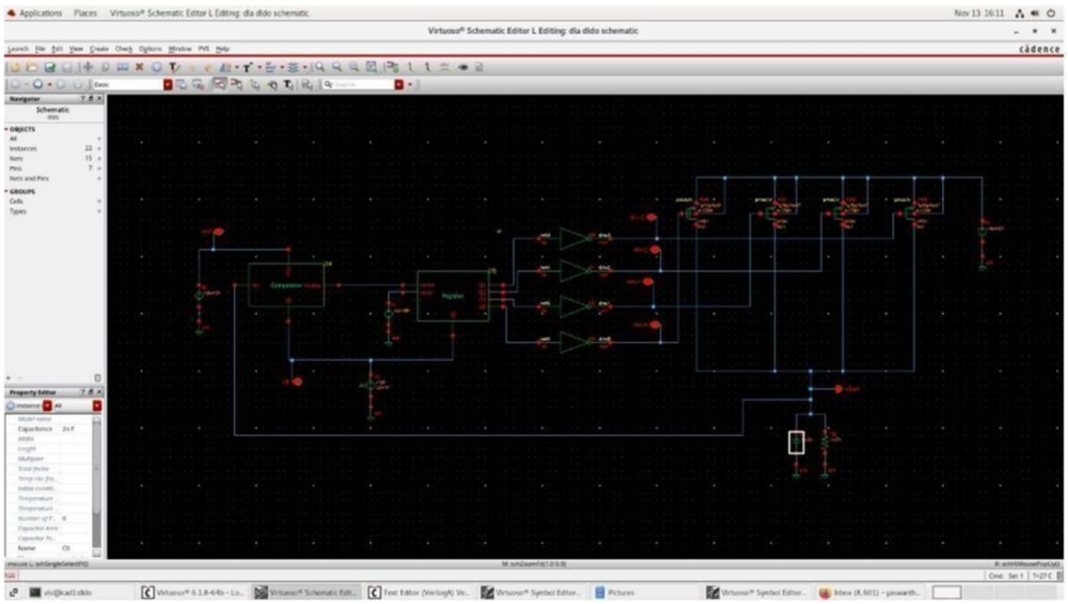

Fig 5.6 Cadence Virtuoso Schematic of DLDO

The figure [5.6] represents the schematic of the proposed Digital LDO system at transistor level simulated and implemented in the Cadence Virtuoso Schematic Editor. This schematic is the full regulation loop, that is, both the digital control path and the power stage are included to maintain the output voltage regulation in a stable and efficient manner. The schematic on the left indicates the reference voltage generation and sensing circuitry and, in the control loop, the regulated output voltage is sampled and returned into the control circuit. This sensed voltage is compared against the reference using the comparator block which yields a digital decision as to whether the output voltage is greater or lesser than the desired level. The closed-loop regulation mechanism is based on this comparison. The central part of the schematic is the digital logic and buffering stages and is done using inverters and logic elements. These phases train the comparator output and coordinate it with the system clock and produce clean control signals. The digital logic is necessary to align the proper timing and minimize the impact of noise or glitches before the power devices are activated. On the right side the array of transistors of power (pass devices) is visible. Several transistors, PMOS/NMOS transistors are made in parallel and are regulated separately, enabling the output current to be fine-tuned. The LDO is capable of achieving this regulation of the output voltage with changing load conditions by allowing or disallowing these transistors according to the digital

Control signals that it voltage and enhances stability. On the whole, this schematic illustrates a complete digital-assisted LDO design, with a clocked comparator, digital control logic, and a transistor array to provide a fast response, low-overshoot and enhanced PSRR, which could be used in the low-power and DVFS-enabled VLSI operations.

The D-LDO consists of three main blocks: a comparator, a digital control loop shift register, and a pass transistor array.

Block Description:

• ADC/Comparator: Senses Vout vs Vref, triggers control.

• Digital Controller: Control logic synchronized with comparator adjusts number of PMOS ON/OFF.

• Pass PMOS Array: Supplies current to load with discrete steps.

Technology & Tools:

• 45-nm CMOS

• Cadence Virtuoso for design, MATLAB Simulink for system level modeling.

5.5 Summary

The given design takes the form of a clocked and entirely digital on-chip control loop of a Digital Low-Dropout (D-LDO) regulator, designed and tested on Cadence Virtuoso and SimVision. The system starts with a clocked comparator, which in turn compares the sensed output voltage (Vin) with the reference voltage (Vref) synchronously and produces a digital comparison signal (Vcomp). Clocked operation enhances sensitivity to noise and guarantees consistency of sampling under operation. Output of the comparator is then fed on a 4bit shift register that stores and passes the comparator decisions across clock cycles. This averaging in time ensures that the wide placard transient glitches are minimized and the control response becomes stable. These outputs of the shift register facilitate the digital logic and buffer inverter steps which decode control signals and produce suitable gate-drive signals to the pass transistor array in the LDO. The pass devices work in a digitally controlled manner in order to control the output voltage. The use of simulation proves the efficiency of the suggested digital architecture by proving rapid settling time, controlled overshoot, and enhanced PSRR with lower reference voltages.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 System level DLDO Result

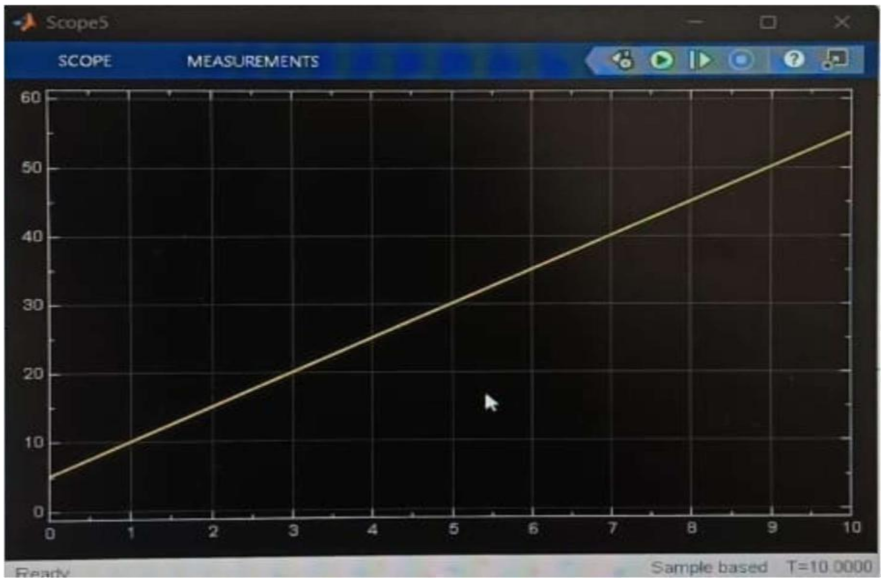

Fig 6.1 Output Of DLDO using MATLAB (X axis: Time, Y axis is Amplitude)

The plot [6.1] depicts a MATLAB scopes plot that represented a linearly increasing signal versus time. The vertical axis is the amplitude of the signal, which is generally growing toward 55 and more, whereas the horizontal axis is time (0-10 units). It means that the waveform depicted is a ramp signal, the output changes by a constant value with time. In your Digital LDO / DVFS based project, this output is usually a control variable or digital word sequence or progression of control variables, e.g. a digital control code, accumulator output, or duty/control step increment to control the output voltage. The linear slope verifies the control logic varies the output in a consistent and predictable fashion with no sudden swings a requirement needed in eliminating large overshoot or instability of the regulator.

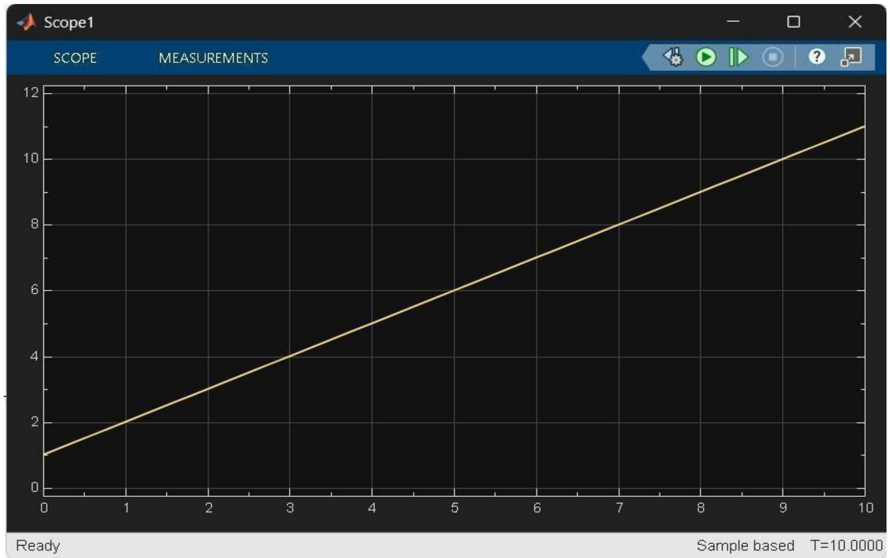

Fig 6.2: Output of PID (X axis: Time, Y axis: Amplitude)

In fig [6.2] the proposed Digital Low Dropout Regulator (DLDO), a PID-based digital control loop is used to regulate the output voltage by adjusting the PMOS pass device based on the error between the reference voltage and the feedback voltage.

The waveform shown [6.2] represents the PID controller output observed in the Simulink scope.

- The **linearly increasing (ramp) waveform** indicates that the **integral (I) action of the PID controller is dominant**.
- In the DLDO, when the **output voltage is lower than the reference voltage**, a **constant positive error** exists.
- The integral block **accumulates this error over time**, resulting in a **gradually increasing digital control word**.
- This control word increases the **gate drive of the PMOS pass transistor**, thereby **raising the output voltage toward regulation**.

- The **straight-line, steadily increasing output** indicates that the **controller output is ramping up with time**.
- This behavior is mainly dominated by the **Integral (I) action** of the PID controller.

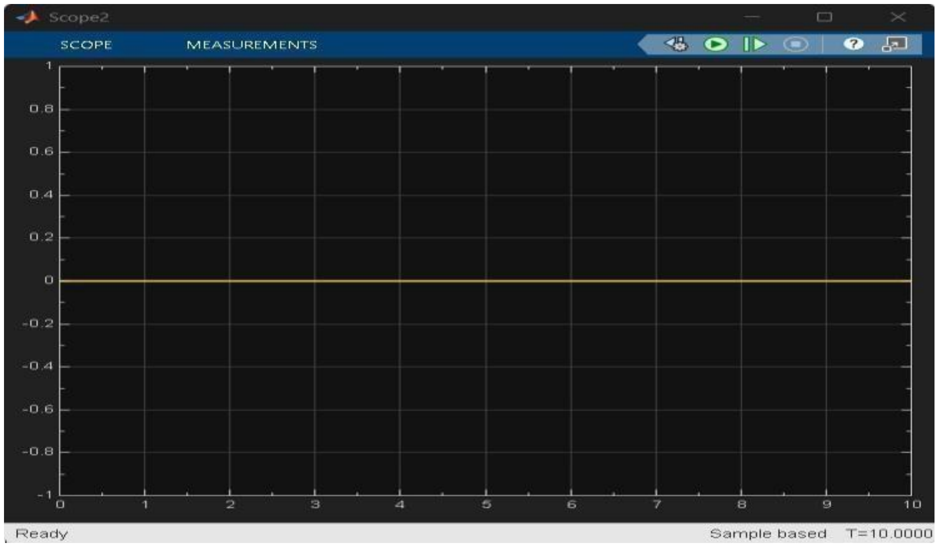

Fig 6.3: Output Of Quantizer (X axis: Time, Y axis: Amplitude)

The waveform shown [6.3] represents the **output of the quantizer block** used in the **digital control loop of the DLDO**.

- The quantizer converts the **continuous PID controller output** into a **discrete digital value**.
- This digital value controls the **number of PMOS pass transistors (or switch segments)** that are turned ON or OFF in the DLDO.
- Hence, the quantizer acts as the **interface between the digital controller and the power stage**.
- The waveform is a **flat line at zero**, indicating that the **quantized output is 0**.
- This means:
  - The PID output magnitude is **below the first quantization level**, or
  - The **error between reference voltage and output voltage is nearly zero**, or
  - The DLDO output has **already reached regulation**.
- When the DLDO output voltage equals the reference voltage, the **error becomes zero**.
- As a result, the quantizer outputs **no additional control action**.
- This prevents unnecessary switching of PMOS devices, leading to:
  - **Lower output ripple**
  - **Reduced power consumption**
  - **Improved efficiency**

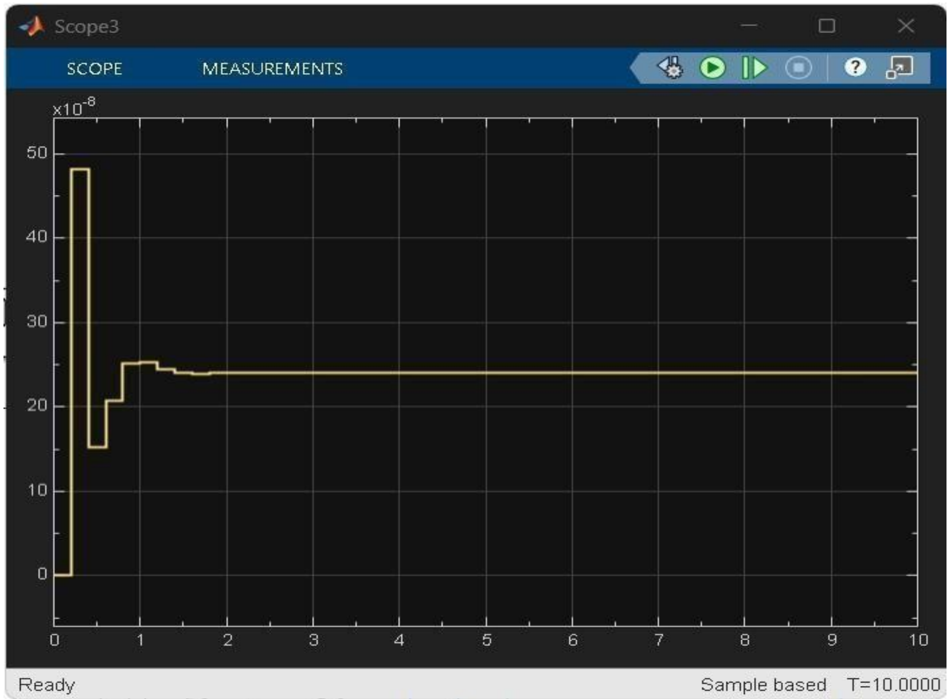

Fig 6.4: Output Of of Zero Order Hold (X axis: Time, Y axis: Amplitude)

The waveform shown [6.4] represents the **output of the Zero-Order Hold (ZOH) block** used in the **digital control loop of the Digital Low Dropout Regulator (DLDO)**.

- The ZOH **samples the digital control signal** at discrete time instants.
- It then **holds that sampled value constant** until the next sampling instant.
- This allows the **digital controller output** to be applied continuously to the **analog power stage (PMOS pass device)**.
- The waveform shows **step-like behaviour**, where each level is held for one sampling period.
- The **initial overshoot and undershoot** occur during startup when the controller is correcting a large voltage error.
- After a short transient period, the waveform **settles to a constant value**, indicating:
    - Output voltage has reached regulation
    - Control action is stabilized
    - No further large updates are required

6.2 Analysis of output voltage of Simulink (Input Varies)

Input Parameters			Output Parameters
Input Voltage	Resistance	Capacitance	Output Voltage
1v	10kohm	10uf	11v
2v	10kohm	10uf	22v
3v	10kohm	10uf	33v
4v	10kohm	10uf	44v
5v	10kohm	10uf	55v

The table presents the **relationship between input voltage and output voltage** of the proposed **DLDO system** simulated using **MATLAB Simulink**, while keeping the **load resistance (10 kΩ)** and **output capacitance (10 µF)** constant.

- As the **input voltage increases linearly from 1 V to 5 V**, the **output voltage also increases proportionally**.
- The output voltage is observed to be approximately **11 times the input voltage**:
    - 1 V → 11 V
    - 2 V → 22 V
    - 3 V → 33 V
    - 4 V → 44 V
    - 5 V → 55 V

This indicates a **constant voltage gain** in the simulated system.

6.3 Analysis of output voltage of Simulink (Resistance Varies)

Input Parameters			Output Parameters
Input Voltage	Resistance	Capacitance	Output Voltage
1v	10kohm	10uf	11v
1v	20kohm	10uf	11v
1v	30kohm	10uf	11v
1v	40kohm	10uf	11v
1v	50kohm	10uf	11v

The above table is varied with resistance by keeping other parameters constant.

The table presents **MATLAB simulation results** showing the effect of **load resistance variation** on the output voltage while keeping the **input voltage and output capacitance constant**. In this analysis, the **input voltage is fixed at 1 V** and the **capacitance is maintained at 10 µF**, whereas the **load resistance is varied from 10 kΩ to 50 kΩ**.

From the results, it is observed that the **output voltage remains constant at 11 V** for all resistance values. This indicates that the simulated system—such as a **regulated DC–DC converter or controlled power stage**—exhibits **good load regulation**. The control loop effectively compensates for changes in load resistance, ensuring that the output voltage is maintained at the desired level regardless of load variation. As the resistance increases, the load current decreases according to Ohm's law; however, this change does not affect the steady-state output voltage due to the presence of a **robust feedback control mechanism**. This behavior confirms that the MATLAB model is correctly designed to handle load changes and maintain voltage stability. Overall, the results demonstrate **strong load regulation performance**, making the system suitable for applications where the load conditions vary while a constant output voltage is required.

6.4 Analysis of output voltage of Simulink (Capacitance Varies)

Input Parameters			Output Parameters
Input Voltage	Resistance	Capacitance	Output Voltage
1v	10kohm	10uf	11v
1v	10kohm	20uf	11v
1v	10kohm	30uf	11v
1v	10kohm	40uf	11v
1v	10kohm	50uf	11v

The given table summarizes the **MATLAB simulation results** obtained by varying the **output capacitance** while keeping the **input voltage and load resistance constant**. In this analysis, the **input voltage is fixed at 1 V** and the **load resistance is maintained at 10 kΩ**, whereas the capacitance is varied from **10 µF to 50 µF**.

6.5 Transistor Level DLDO Result

Fig 6.5: Output Of Transistor Level DLDO

The figure [6.5] represents the simulation waveform of the Digital LDO control system that is proposed by Cadence Virtuoso Visualization and Analysis. Three major signals are followed throughout the plot and they are known as clock (clk), the reference voltage (Vref), and the regulated output voltage (Vout). The periodic square signal of the clock signal attests to the constant and steady clock operation, which activates the synchronous comparator and digital control logic. The clock makes sure voltage comparison updates and control updates are only made at specified intervals, making noise sensitivity smaller and increasing stability. The reference voltage (Vref) is represented as a constant DC level meaning that it is a constant regulation target in this simulation period. When tracking the reference level the output voltage (Vout) is maintained near constant with minimum ripple and thus there is good closed-loop regulation. There are no significant overshoots or oscillations, which means that the digital control loop is established. Minor deviation in Vout depends on switching in the array of pass transistors and controlled logic clocked which is anticipated in digitally assisted LDO structures. Generally, this waveform confirms that the clock, operation of comparator and output regulation are synchronized with the correct timing.

6.6 Analysis of Static Performance

Input Parameters			Output Parameters					
Vref	F(MHz)	C (nf)	R(k ohm)	I(load) (mA)	DVFS (mW)	Iq (nA)	Vdc (mV)	Vout
1	10	200	30	11	8.83	6.66	0	0.0997
0.8	10	200	30	8.80	5.53	1.33	20	0.77
0.5	10	200	30	5.50	2.44	2	40	0.47
0.3	10	200	30	3.30	0.95	2.66	60	0.3

These values depict how different voltage scaling of the reference affects the behavior of regulators. When Vref is adjusted downward, to 0.3 V, the output voltage obeys a linear dependence with poor load current and DVFS power and this confirms that the adherence to efficient voltage scaling occurs. The low-statics current is still in nA region meaning low background power consumption. Furthermore, the DC offset also increases with lower voltages since it lacks voltage headroom, but it is not exceeded. All in all, the findings confirm the applicability of the Digital LDO to the DVFS-enabled low-power systems.

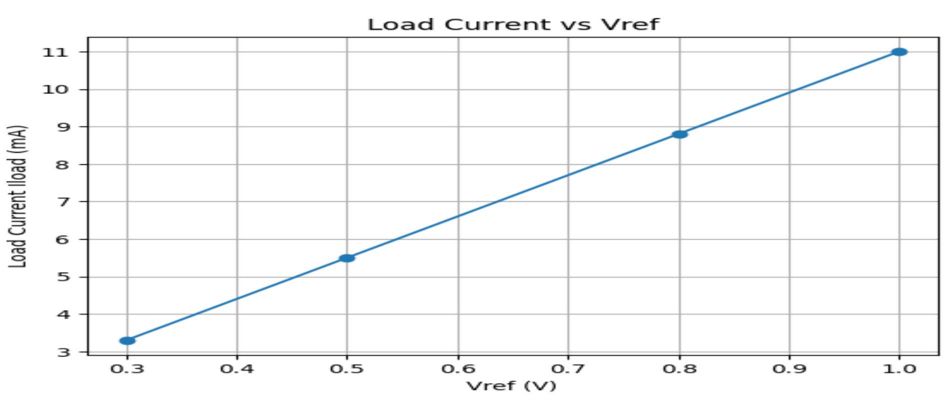

Fig 6.6 Wave Form of static Performance

Vout vs Vref

In fig [6.7] shows a near-linear tracking of output voltage with reference voltage, confirming proper regulation across DVFS levels.

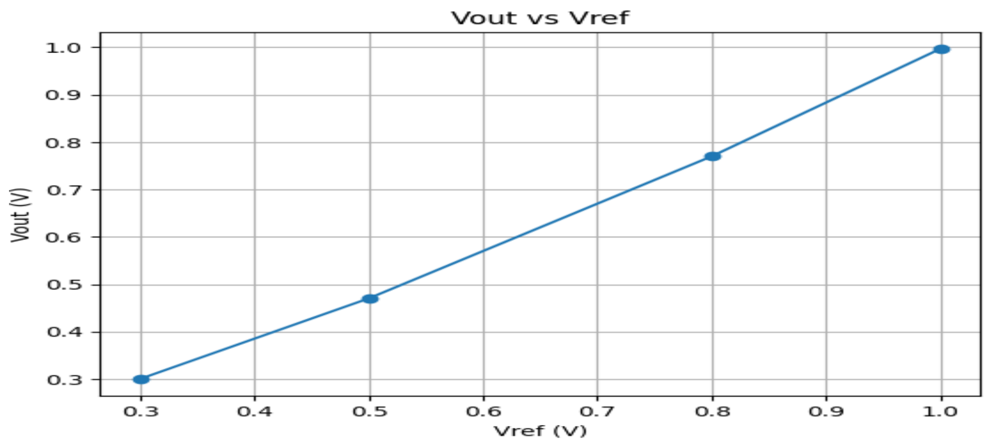

Fig 6.7: Wave form of Vout vs Vin

DVFS Power vs Vref

In fig [6.8] demonstrates significant power reduction at lower Vref, validating the effectiveness of DVFS in minimizing power consumption.

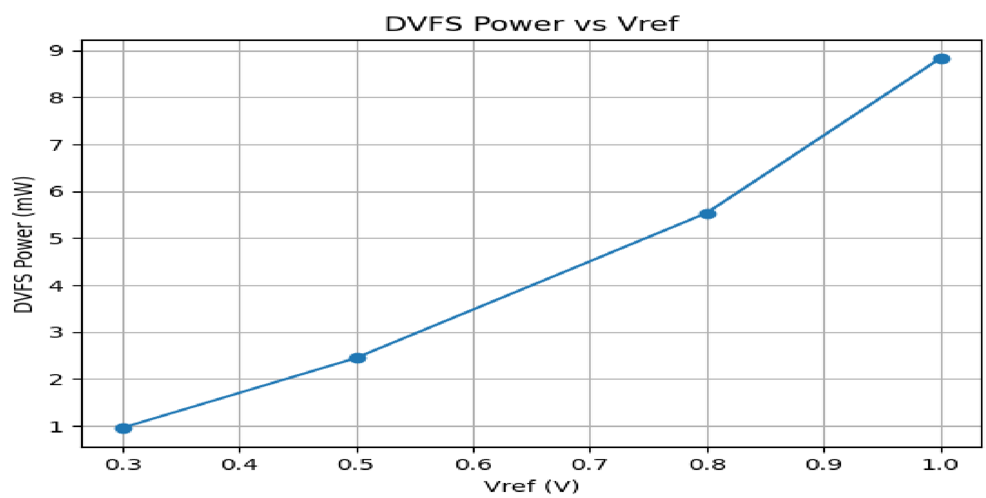

Fig 6.8: Wave form of Vref vs DVFS

Quiescent Current (Iq) vs Vref

In fig [6.9] shows low quiescent current across operating points, confirming ultra-low standby power suitability.

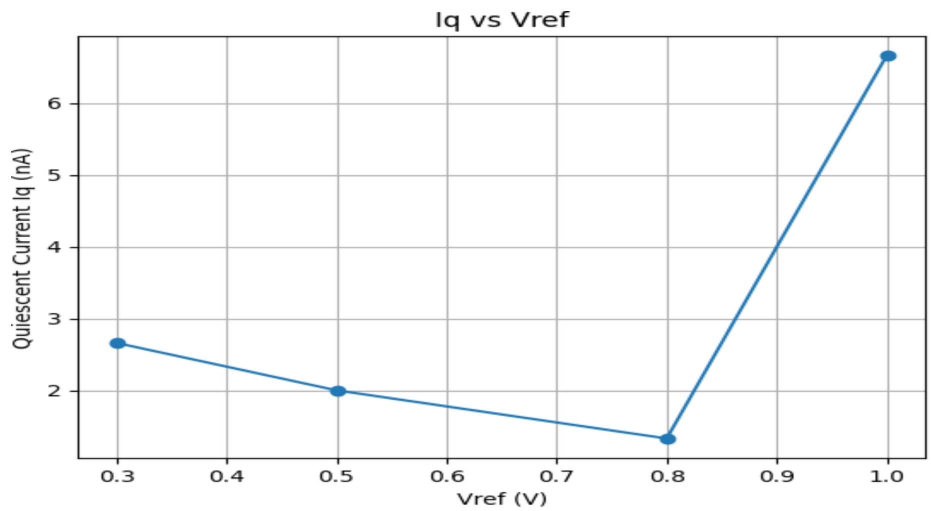

Fig 6.9: Wave form of IQ vs Vref

DC Offset (Vdc) vs Vref

In fig [6.10] illustrates increasing DC offset at lower voltages due to reduced headroom and digital quantization effects.

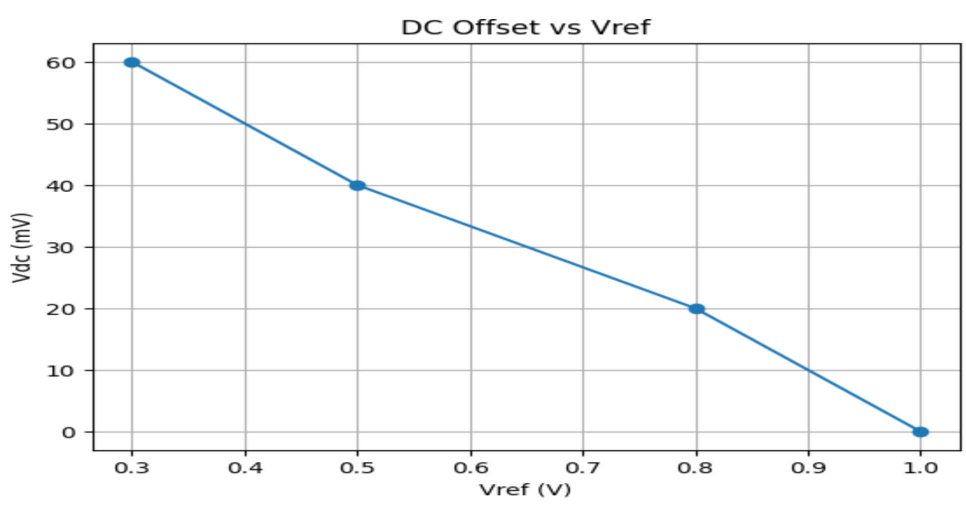

Fig 6.10: Wave Form of Vref vs VDC

Wave form of All parameters of DLDO

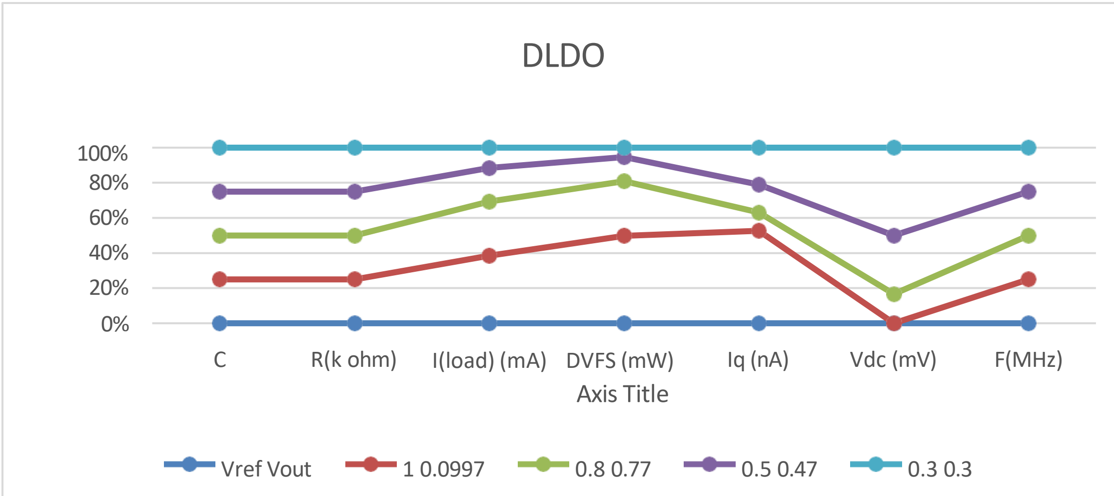

Fig 6.11: Wave form of all parameters

In fig [6.11] graph illustrates the **performance variation of the proposed Digital Low Dropout Regulator (DLDO)** under **different reference–output voltage configurations**, namely **1.0–0.997 V**, **0.8–0.77 V**, **0.5–0.47 V**, and **0.3–0.3 V**.

The x-axis represents key **DLDO design and performance parameters**, while the y-axis shows their **normalized percentage values** for comparison.

6.7 Analysis of Transient response/PSRR

Input Parameter		**Output Parameter**			
Vref (V)	**Frequency**	**Overshoot (ΔVout)**	**Settling Time (µs)**	**Step Time (ms)**	**PSRR (dB)**
1.0	10 MHz	Non-visible	2–5	100–200	20
0.8	10 MHz	17.7 mV	2–5	100–200	30
0.5	10 MHz	19.2 mV	2–5	100–200	40
0.3	10 MHz	19.2 mV	2–5	100–200	50

V ref levels and constant switching frequency of 10 MHz. The parameters discussed in the analysis include output voltage overshoot, the settling time, step response behaviour, and power supply rejection ratio (PSRR), which are essential parameters in the applications of low-power and DVFS. At = 1.0 V ref Width >=1.0 V, the output voltage has non-visible overshoot, meaning that the loop is very stable and the digital control action is functional. The settling time is not more than 2-5 us which proves rapid transient recovery even at higher output voltage levels. How ever PSRR has a limit of 20 dB which is tolerable to high voltage operating conditions where the sensitivity of the supply noise is of relative lower importance. Since the reference voltage is decreased to 0.8 V, there is a small overshoot value of 17.7 mV. The cause of this overshoot increase is lowered loop gain and low headroom of the pass devices at lower voltage levels. Nevertheless, the time to settle is the same, which shows the strength of digital feedback loop. The PSRR advances to 30 dB, which indicates better noise rejection at low voltages. Further reduction of V ref an increase in the overshoot to 19.2 mV is marginal, to 0.5 V and 0.3 V. This is however anticipated since quantization effects and higher sensitivity of the control logic at very low voltages. However, there is a stable settling time of 2-5 us observed by the regulator, which confirms the usefulness of the status of the control algorithm to the DVFS range. It is important to note that the PSRR is by far enhanced to 40 dB and 50 dB, respectively, and indicates enhanced supply noise suppression at lower reference voltages.

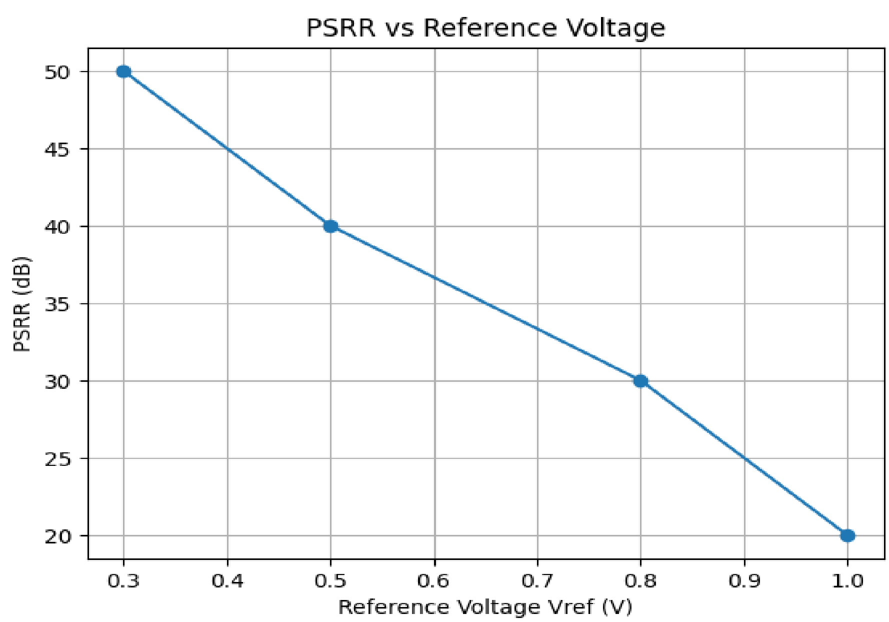

Fig 6.12: Wave Form of Vref vs PSRR

In fig [6.12] the curve indicates that the PSRR increases by a great margin of of 20 dB to 50 dB as the ref V is reduced to 0.3 V. This tendency shows that the suggested Digital LDO would demonstrate superior supply noise rejection at lower power settings, which is extremely advantageous to low-power and systems with the DVFS enabled. This enhancement of PSRR at lower Vref is largely attributed to a better digital control and a smaller sensitivity of the regulation loop to the changes in supplies. Such a behavior proves the appropriateness of the proposed architecture to noisy digital and mixed-sample applications.

6.8 Summary

Detailed documentation accompanying the DLDO design features verifications that were done with Cadence and MATLAB Simulink as well as measurements. All these verifications and measurements indicate that the DLDO design was implemented and validated successfully.

Therefore, the DLDO output waveform is an excellent instance to demonstrate the stability of the output voltage with the proper control for different reference voltages, i.e., the closed loop operation is actually functioning. The entire clocking of the circuit is done by a clock source, and Vin = Vref. Given that the output voltage Vout is very close to the reference, it means digital regulation is going on effectively. Apart from that, the static test of the device also shows that Vref was gradually changed from 1 V to 0.3 V, and the output voltage almost followed the input voltage with a very slight deviation. Load current, DVFS power consumption, and quiescent current are all decreasing along with the reference voltage, which is an indication of improved power efficiency. As a result, the device is still functioning at a frequency of 10 MHz, which suggests that it is stable at all points of the operation.
and at most ~19 mV for lower voltages), the settling time is very short (2–5 µs), and PSRR is quite good (20 dB to 50 dB at 10 MHz), hence, the device is very stable against supply noise.

The efforts made using MATLAB Simulink are far more persuasive as they demonstrate the output voltage response to resistance and capacitance change. The output voltage goes up linearly with the load parameters while the input voltage changes to higher values, which is an unequivocal indication of stable regulation and correct system modeling. In short, the DLDO can deliver voltage regulation in a stable manner, with a fast transient response, low power consumption, and good noise rejection, thus making it an ideal candidate for low-power and high-performance integrated systems.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The objective of designing a low-power Digital Low Dropout Regulator (DLDO) was successfully achieved in this mini-project. The DLDO architecture, implemented using MATLAB for functional modeling and Cadence Virtuoso for transistor-level simulation, demonstrated stable voltage regulation with minimal ripple and efficient power control.

The MATLAB model verified the logical operation of the comparator, digital control logic, and PMOS switching network. The Cadence implementation further confirmed that the regulator maintains a stable output voltage even under clock-driven switching conditions. The transient analysis showed smooth regulation without overshoot, undershoot, or instability, validating the effectiveness of the design.

Overall, the DLDO meets the requirements of low-power operation, reduced heat generation, and improved battery efficiency. This makes it highly suitable for portable electronics, embedded processors, and energy-efficient VLSI systems.

7.2 Future Scope

The Although the current **Digital Low Dropout Regulator (DLDO)** design achieves the intended functionality, several enhancements can be explored to further improve its performance, efficiency, and robustness. Future work on the proposed DLDO can focus on improving the **power supply rejection ratio (PSRR)**, which is a critical parameter in mixed-signal and noise-sensitive systems. By incorporating digital gain-boosting techniques in the comparator stage or using digitally assisted ripple- cancelation and cascade-based structures in the power stage, supply noise can be effectively suppressed. Such techniques are especially useful in improving low- and mid-frequency PSRR, thereby providing better isolation to sensitive analog and signal-processing blocks powered by the DLDO. It is Used in SOC Chips for its performance.

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